



UNIVERSITY
Of KIGALI

"Unequalled Education Excellence"

CONTINUOUS ASSESSMENT TEST

SCHOOL OF COMPUTING AND INFORMATION TECHNOLOGY

PROGRAMME

Bachelor degree with honours in computer science, Bachelor degree with honours in information technology and Bachelor degree with honours in business information technology

MODULE CODE AND TITLE:
CIT83104: ADVANCED WEB DESIGN AND DEVELOPMENT

Year 3, TRIMESTER 1

DATE:
19-04-2026

TIME:
9:30AM-12:30PM
SESSION: WEEKEND

DURATION:
3 HOURS

TOTAL MARKS:
40

EXAMINER/ LECTURER:
Mr. Jean Aime NDABAHARIYE

MODERATORS:
Dr. Nteziriza Nkera Bahizi Josbert
Mr. MANIRARORA JEAN MARIE VIANEY

INSTRUCTIONS TO THE CANDIDATE

1. All questions must be attempted in the answer booklet provided.
2. All queries should be directed to the invigilator; do not communicate or attempt to communicate with any other candidate.
3. You have **THREE HOURS** to complete this paper. You are not allowed to leave the examination room within the **first hour** and in the last 15 minutes of this examination.
4. This is a **CLOSED BOOK** examination.
5. Answer 4 Questions in all, Question One in each section is Compulsory.
6. Choose any 1 other question in from Section A.
7. Choose any 1 other question in from Section B.
8. Please, do not write on the question paper.

SECTION A

Question 1 - Compulsory

An institution wants to develop a web-based student registration system using PHP and MySQL to improve data management. The system should allow the registrar to enter student details such as full name, registration number, email, and programme through an HTML form. Once the data is submitted, it should be stored in a database for future reference. The system must also allow viewing all registered students in a structured table format. Additionally, the system should support joining related tables and redirecting users after completing actions.

Tables:

students (student_id, full_name, reg_no, email, programme_id) programmes
(programme_id, programme_name)

- Write PHP code to establish a database connection using MySQLi. (3 Marks)
- Write **query** to retrieve records by joining the students and programmes tables and display the results. (4 Marks)
- Write PHP code to redirect the user to another page after 5 seconds of performing an action (e.g., after inserting data, open **view.php** page). (3 Marks)

Question 2

A company maintains employee records in a MySQL database to manage staff information efficiently. The HR manager wants to update employee details such as name, department, salary, and phone number when changes occur. The system should allow selecting an employee using their ID and display the existing data in a form. After editing, the updated information should be saved back into the database. The updated records should then be displayed to confirm the changes and allow to print the employee's information. Database is employee_db

Table: employees (emp_id, emp_name, department, salary, phone)

- Write PHP code to fetch employee data by emp_id and display in a form. (3 Marks)
- Write PHP code to update employee details. (4 Marks)
- JavaScript to print out the unique employee's identification. (3 Marks)

Question 3

BERWA Shop wants to develop a simple web-based invoice generation system using PHP and JavaScript without using a database. The system should allow the cashier to enter client name, product name, quantity, unit price and tax rate. The system should use JavaScript to validate inputs and perform simple calculations and PHP to process and display the final invoice report. The system should also allow the cashier to print or download the invoice.

Questions

- a) Design an HTML form to capture client and product details (2 Marks)
- b) Use JavaScript to validate the form and ensure all fields are filled and numeric values are correct (2 Marks)
- c) Use JavaScript to calculate subtotal tax and total amount automatically (3 Marks)
- d) Write PHP code to process the data and display the invoice report (2 Marks)
- e) Use JavaScript to add a button for printing or downloading the invoice (1 Marks)

SECTION B

Question 4 - Compulsory

A small business wants to build a product inventory system using PHP and MySQL. The system should allow users to add new products including product name, price, and quantity. It should also display all products in a table for monitoring stock levels. Additionally, users should be able to update product details and delete products when necessary and add option of searching by product_name. This system will improve stock management and reduce manual errors. Database name is stock_db.

Table: products (product_id, product_name, price, quantity)

- a) Write PHP code to insert a product. (2 Marks)
- b) Write PHP code to display all products and search. (4 Marks)
- c) Write PHP code to update and delete a product. (4 Marks)

Question 5

A college wants to develop a web-based marks management system using PHP and MySQL. The system should allow the trainer to enter student details and marks for CAT, and exams and allow the trainer to select the module at least two modules be available into the modules' table. After submission, the system should automatically calculate the total marks and assign a grade based on the grading policy. All records should be stored in the database and displayed in a report table. This system will simplify academic performance tracking, and database named as academic_db.

Tables:

marks (marks_id, student_name, reg_no, cat_60, exam_40, total, grade, module_id)

modules (module_id, module_name, module_code)

- a) Write HTML form to capture module, students' details and marks. **(3 Marks)**
- b) Write PHP code to calculate total and assign grade. **(3 Marks)**
- c) Write PHP code to insert and display student results. **(4 Marks)**

Question 6

A web application requires a secure user authentication system using PHP and MySQL. The system should allow users to register by entering their full name, email, country, gender, and password through an HTML form. The password must be securely encrypted before being stored in the database. After registration, users should be able to log in using their email and password. Upon successful login, a session should be created to maintain user access, and users should be able to log out securely.

Table: users (user_id, full_name, email, country, gender, password)

- a) Write HTML form using appropriate input types (text, email, select, radio, password) to capture full name, email, country, gender, and password. **(2 Marks)**
- b) Write PHP code to encrypt the password before storing it. **(2 Marks)**
- c) Write PHP code to insert user data into the database. **(2 Marks)**
- d) Write PHP code for user login using email and password. **(2 Marks)**
- e) Write PHP code to create and destroy a session (login & logout). **(2 Marks)**

END